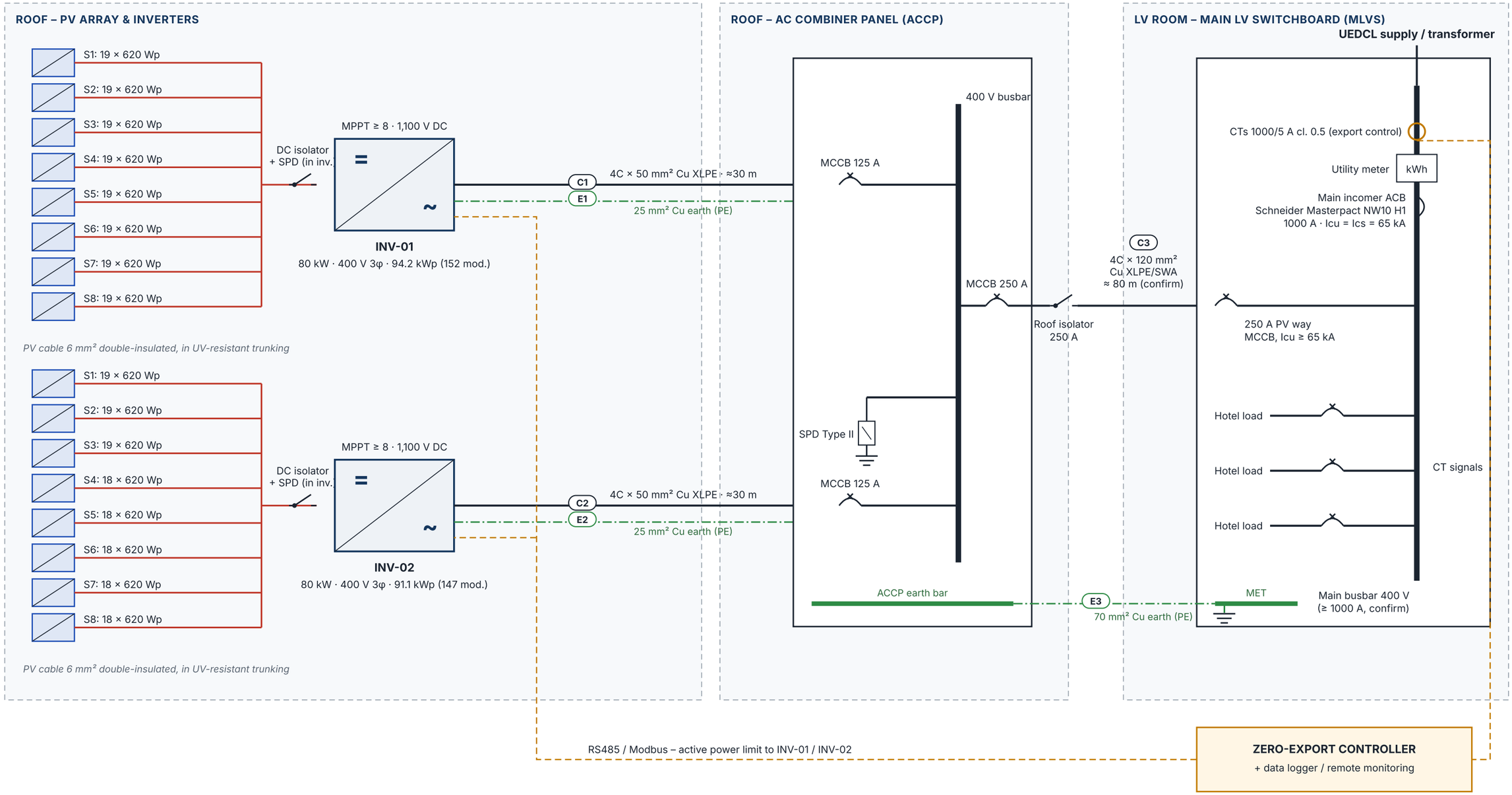
SINGLE-LINE DIAGRAM – 185.4 kWp ROOFTOP SOLAR PV SYSTEM (GRID-TIED, ZERO EXPORT, NO BATTERY)

Kampala Marriott Hotel – mezzanine roof · 299 × 620 Wp modules · 2 × 80 kW string inverters



EARTHING & LIGHTNING PROTECTION

Array frames & rails bonded (6/16 mm² Cu) → main earth bar → building earth
Integrated with existing lightning protection system (IEC 62305); SPDs on DC and AC sides

LEGEND

DC (PV) · AC 400 V · Control / comms · Circuit breaker · Isolator · Earth (PE)

Strings: INV-01 8 × 19 · INV-02 3 × 19 + 5 × 18 · DC/AC 1.16 · Indicative – confirm with final equipment datasheets

CABLE SCHEDULE

Tag	From → To	Cable	Length	Design current	Protection	Current rating (derated)*	Voltage drop	Sizing basis
S1–S16	PV strings → inverter DC inputs	2 × 6 mm² Cu H1Z2Z2-K	≈ 40 m	20.1 A (Isc × 1.25)	Inverter DC isolator + SPD	≈ 70 A	≈ 0.6 %	IEC 62548 string current; 1,500 V DC solar cable
C1, C2	INV-01 / INV-02 → ACCP	4C × 50 mm² Cu XLPE	≈ 30 m each	115.5 A	MCCB 125 A (ACCP)	≈ 168 A	0.74 %	Ib ≤ In ≤ Iz: 115.5 ≤ 125 ≤ 168 A (hot roof, 2 circuits grouped)
C3	ACCP → MLVS 250 A PV way	4C × 120 mm² Cu XLPE/SWA	≈ 80 m	231 A	MCCB 250 A + isolator	≈ 330 A	1.57 %	Ib ≤ In ≤ Iz: 231 ≤ 250 ≤ 330 A; total AC drop 2.3 % (< 3 %); armoured for riser route
E1, E2	INV-01 / INV-02 → ACCP earth bar	25 mm² Cu (green/yellow)	≈ 30 m each	–	–	–	–	IEC 60364-5-54: line > 35 mm² → PE ≥ S/2 = 25 mm²
E3	ACCP earth bar → MET (LV room)	70 mm² Cu (green/yellow)	≈ 80 m	–	–	–	–	IEC 60364-5-54: 120 / 2 = 60 → next size 70 mm²; armour earthed both ends

** Typical IEC 60364-5-52 ratings (XLPE, Cu, on perforated tray / in air) derated for 45 °C roof ambient and grouping. Confirm against the cable manufacturer's data and the final route.*